

JINGHAN ZHENG

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EDUCATION

McGill University

Montreal, Canada

B.Sc. **Honours** Computer Science | CGPA: 3.77/4.00 (First class honours)

Expected Jun 2027

Selected coursework: Natural Language Processing (A), Applied Machine Learning (A), Theory of Computation (A), Statistics (A), Probability (A-), Numerical Computing (A-), Honours Algorithms and Data Structures, Programming Languages and Paradigms.

Planned coursework: Artificial Intelligence, Honours Algorithm Design, Reinforcement Learning, and From Natural Language to Data Science.

RESEARCH INTERESTS

Natural language processing; Large language models; Contextual Robustness and In-Context Learning.
LLM reliability and evaluation; Mechanistic interpretability

PUBLICATIONS

1. From Relation-Specific Heads to Shared Interventions: Contextual Entrainment in Language Models.

First author, Accepted by AACL-IJCNLP 2026 Main Conference.

Experiment artifact: github.com/billyzheng2410/context-related-hallucination

2. From Answer Voting to Claim Auditing: Objective-Aware Hallucination Control for RAG under Corrupted Evidence

First author, Under review at AAAI 2027.

RESEARCH EXPERIENCE

Contextual Entrainment in Language Models

Montreal, Canada

McGill University

Researcher | First Author

2026-Present

Investigated whether attention-head interventions for contextual entrainment are relation-specific or contain reusable components shared across semantic relations.

Built a factual-distractor benchmark spanning 19 lexical-semantic subtypes and four semantic families to measure graded transfer across same-subtype, same-family, and different-family conditions.

Applied differentiable Gumbel-Sigmoid head search and causal ablation to identify anti-entrainment interventions in LLaMA-3.1-8B.

Analyzed global, family-residual, and subtype-residual components through transfer, overlap, and additive intervention tests; found a reusable global core with semantic modulation.

Replicated the main transfer pattern on Mistral-7B and Qwen2.5-7B and evaluated clean relation-completion and in-context-use side effects.

OR-Wolf: Claim-Level Auditing for Reliable Multi-Agent RAG

Independent Research | First Author

Developed OR-Wolf, a multi-agent RAG framework that audits atomic claims against reference evidence to control unsupported content under corrupted retrieval.

Designed objective-aware claim retention and answer finalization to balance wrong answers, false abstentions, deletion of supported evidence, and audit cost.

Evaluated three Qwen2.5-7B-Instruct agents with a DeBERTa-v3 verifier on SciFact; compared answer voting, agent filtering, and claim auditing with matched ablations.

On 200 SciFact examples with one or two corrupted agents, improved accuracy by 6-13 percentage points over standard multi-agent debate and reduced wrong non-abstaining predictions by 77.6-80.7% relative, with increased abstention.

Agricultural Multi-Task Vision Foundation Model

Nanjing Agricultural University & Huazhong University of Science and Technology

Remote

Research Assistant | Supervisors: Prof. Shouyang Liu and Prof. Hao Lu

2026-Present

Developed a DINOv3-based agricultural multi-task vision model for wheat disease classification, wheat-head detection, and organ segmentation across class_disease, GWHD, and GWFSS datasets.

Extended MTLoRA with shared and task-specific low-rank adapters injected into ViT qkv/projection layers, plus decoupled MLP, Faster R-CNN, and FCN heads for task-specific prediction.

Implemented multi-task training, inference, and evaluation pipelines in PyTorch/timm with randomized task sampling, AdamW optimization, checkpointing, and task-specific metrics.

Compared MTLoRA/HPS baselines with Align-MTL variants using MK-MMD feature-distribution alignment; identified faster convergence but a segmentation alignment tax, motivating hybrid alignment plus hard-decoupling designs.

LLM-Based Multimodal Psychosocial Analysis for the AI-Digital Nurse Avatar Shared Reality Lab, McGill University

Montreal, Canada

Undergraduate Research Assistant | Supervisor: Prof. Jeremy Cooperstock
Built and deployed a Dockerized multi-service platform with a FastAPI backend, React frontend, Traefik reverse proxy, Whisper ASR, and Weaviate-based semantic retrieval.

Apr-Jul 2025

Developed an end-to-end multimodal psychosocial analysis pipeline combining Word2Vec transcript embeddings, audio spectrogram features, and a TensorFlow/Keras CNN-Highway-LSTM model.
Implemented session-level depression-risk scoring and EWMA-based anomaly alerts for longitudinal monitoring in the AI-Digital Nurse Avatar (ADiNA) system.

Diagnosed model-loading, feature-shape, and inference-consistency issues across audio and text modalities and integrated the corrected pipeline into the deployed system.

AI-Assisted Tactile Chinese Character Rendering System

Alibaba-Zhejiang University Joint Institute of Frontier Technologies (EAGLE LAB)

Hangzhou, China

Research Intern | Advised by Prof. Jiajun Bu

May-Jul 2024

Contributed to an AI-assisted writing system for visually impaired users by developing software modules for tactile character rendering and hardware control.

Designed a pipeline that converted Chinese characters into dot-matrix representations and generated corresponding motor-control commands.

Implemented PWM-based control for a matrix of stepper motors and integrated the module into the main accessibility-oriented prototype.

SELECTED TECHNICAL PROJECTS

Scientific Text Classification: LSTM vs. BERT | Python, PyTorch, Hugging Face Transformers

Implemented a multi-layer LSTM classifier and fine-tuned BERT on WOS-11967 for hierarchical scientific-text classification; compared validation performance across label levels.

Analyzed attention patterns for correct and incorrect predictions and conducted ablations on dropout, batch normalization, activations, and forget-gate bias.

Fashion-MNIST Classification: MLP, CNN, and Transfer Learning | Python, PyTorch

Built an MLP training pipeline from scratch, including forward propagation and backpropagation, and evaluated depth, activation, normalization, and regularization choices.

Achieved 86.71% test accuracy with the best MLP and 92.16% with a CNN; compared performance with frozen ResNet-18 transfer learning.

Model Selection and Regularization | Python, NumPy, scikit-learn

Modeled nonlinear data with Gaussian basis functions, used cross-validation to select model complexity, and analyzed underfitting and overfitting.

Implemented ISTA for L1-regularized optimization and compared L1 and L2 regularization in terms of sparsity and generalization.

Linear and Logistic Regression with Mini-batch SGD | Python, NumPy, scikit-learn

Implemented analytical linear regression and mini-batch SGD for Parkinson telemonitoring prediction and built a logistic classifier for breast-cancer diagnosis.

Evaluated optimization, regularization, and generalization behavior across regression and classification settings.

Android Music Player Application

Team Leader | Java, Android Studio

Shenyang, China

Jul-Aug 2023

Led development of an Android music application with authentication, playback, volume control, search, and download functionality.

Integrated an online database and implemented music organization by artist, album, and genre.

Code portfolio: github.com/billyzheng2410/My-CS-course-work

LEADERSHIP AND AWARDS

McGill Student Chinese Debate Club

President (2025-2026); Vice President (2023-2025)

Montreal, Canada

Sep 2023-Present

Led recruitment, training, tournament preparation, and collaboration with debate teams at universities across Canada.

Champion and Grand Final Best Debater, 2025 Canada National Chinese Debate Competition; team runner-up at the 2023 and 2024 Chinese Debate World Cup Canada Regional.

TI-Nspire Handheld Technology Innovation Competition

National First Prize | Team project on automatic compilation of Python code for graphing calculators

China

GameDev McGill

Club Member | Unity game development

Montreal, Canada

Sep 2023-Present

Contributed to a 3D strategy-game project and participated in technical discussions and peer knowledge sharing.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, R, MATLAB.

Machine learning and NLP: PyTorch, TensorFlow/Keras, Hugging Face Transformers, TransformerLens, timm, torchvision, DINOv3, LoRA/MTLoRA, Faster R-CNN, FCN segmentation, scikit-learn, NumPy, pandas, Word2Vec, Whisper.

Systems and development: Git, Linux, Docker, FastAPI, React, Traefik, Weaviate, Jupyter.

Languages: Mandarin Chinese (native), English (full professional proficiency).